



# Unitary operators preserve the inner product between the vectors they act on.

Asked 4 years, 11 months ago   Modified 4 years, 11 months ago   Viewed 4k times



Let  $H$  be a Hilbert space and  $U$  a unitary operator on  $H$ , that is,  $U^*U = UU^* = I$  where  $U^*$  denotes the adjoint of  $U$ . Then it is immediately clear that for  $f, g \in H$  we have

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$$\langle Uf, Ug \rangle = \langle f, U^*Ug \rangle = \langle f, g \rangle.$$



However in the physics text I am reading, the proof is as follows:



Let  $|V_1'\rangle = U|V_1\rangle$  and  $|V_2'\rangle = U|V_2\rangle$  be kets. Then

$$\langle V_2'|V_1'\rangle = \langle UV_2|UV_1\rangle = \langle V_2|U^*U|V_1\rangle = \langle V_2|V_1\rangle.$$

I am confused by this notation. Wouldn't  $\langle V_2'|V_1'\rangle$  denote a (bra, ket) pair? Or in this case is it simply the inner product of  $V_2'$  with  $V_1'$ ? Moreover, what is the meaning of  $\langle V_2|U^*U|V_1\rangle$ ? I get that of course  $U^*U = I$  but the double  $\langle \cdot | \cdot \rangle$  notation is throwing me off.

Another example of the double  $|$  notation:

$$\delta_{ij} = \langle i|U^*U|j\rangle,$$

where  $\delta_{ij} = 1_{\{i=j\}}(i, j)$  is the Kronecker delta. What is the meaning of  $\langle i|U^*U|j\rangle$ ?

notation   hilbert-spaces   physics   inner-products

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edited Jan 1, 2020 at 20:46



J.G.

117k

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asked Jan 1, 2020 at 20:35



Math1000

37.7k

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Note: I am posting this here instead of [physics.stackexchange.com](https://physics.stackexchange.com) because fundamentally this is a question about mathematical notation. – Math1000 Jan 1, 2020 at 20:36

1 Answer

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In the bra-ket notation,  $\langle x|y\rangle$  means the scalar product  $\langle x, y\rangle$  of  $x$  and  $y$ , and  $\langle x|A|y\rangle$  means the scalar product  $\langle x, Ay\rangle$  (here it's convenient to use the convention that the scalar product is linear in the second and antilinear in the first variable). Thus, the meaning of  $\langle V_2|U^*U|V_1\rangle$  is just  $\langle V_2, U^*UV_1\rangle$  which is equal to  $\langle V_2, V_1\rangle$  as  $U^*U = \text{id}$ .

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answered Jan 1, 2020 at 20:41

Vadim Alekseev

1,363 6 13

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But I thought a bra-ket  $\langle\varphi|\psi\rangle$  meant that  $\langle\varphi| = f_\varphi$  denoted an element in the dual space, by  $f_\varphi(\psi) = \langle\varphi|\psi\rangle$  for  $\psi \in H$ . – [Math1000](#) Jan 1, 2020 at 20:58

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This is an equivalent definition: by the Riesz representation theorem there is a canonical *antilinear* isomorphism between the dual space of a Hilbert space and the Hilbert space itself. – [Vadim Alekseev](#) Jan 1, 2020 at 21:01

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Ah yes, of course. Thank you. – [Math1000](#) Jan 1, 2020 at 21:04

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